

Shelton Bumhe

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Education

University of Nebraska–Lincoln

B.S. Computer Science, Minor in Mathematics

Expected Dec 2027

GPA: 3.71/4.00

Technical Skills

Languages: Java, Python, C/C++, C#, JavaScript, SQL

Backend & Distributed Systems: Spring Boot, REST APIs, Kafka, Redis, Microservices, Docker, Low-Latency Systems

Testing & Automation: JUnit, JMeter, Postman, CI/CD, Azure DevOps, LangGraph, MCP, Generative AI

Professional Experience

Fiserv

Omaha, NE

Software Engineer Intern

May 2026 – Aug 2026

- Engineered and debugged Java/Spring Boot microservices processing 2M+ Kafka events/day across 8+ enterprise systems, containerizing services with Docker and integrating real-time workflows with legacy COBOL systems.
- Used Oracle for persistent state and Redis for low-latency access, delivering sub-150 ms file-status searches and reducing production investigations from 3+ hours to 3 seconds.
- Built asynchronous Python validation services with LangGraph, MCP, and Pydantic to detect stalled processing and enforce business rules, supporting \$50K/month in projected SLA savings.

Ameritas

Lincoln, NE

Software Engineer Intern

Aug 2026 – Present

- Engineered integration workflows connecting 3+ third-party platforms with internal enterprise services, automating data ingestion, transformation, validation, and routing across system boundaries.
- Developed backend integration logic to normalize package and transaction payloads across external and internal services, reducing manual handoffs in enterprise processing workflows.
- Implemented logging, error handling, and reconciliation across 10+ integration points to detect malformed payloads, failed transactions, and downstream processing issues before propagation.

Nebraska Department of Health and Human Services

Lincoln, NE

Data Engineering Intern

Aug 2026 – Present

- Automate legacy COBOL reporting workflows by rebuilding business logic in Azure Databricks, PySpark, Delta Lake, and Snowflake, transforming 1M+ records into standardized cloud datasets.
- Optimize manual reporting and validation through reusable ETL pipelines, source-to-target transformations, automated reconciliation, and data-quality checks for missing, duplicate, and mismatched records.
- Tested, debugged, and shipped 15+ approved changes through Azure DevOps CI/CD across Databricks and Snowflake DEV/UAT environments.

Projects

GrandPilot — AI Grant Search Platform | Python, FastAPI, LangGraph, MongoDB, Redis Apr 2026 – Present

- Built an AI-powered grant discovery platform that helps founders search, rank, and evaluate relevant government funding opportunities based on company profile and eligibility.
- Developed a Corrective RAG pipeline with LangGraph, MongoDB Atlas Vector Search, hybrid semantic/keyword retrieval, reranking, and source-grounded eligibility validation.
- Implemented Redis caching, rate limiting, and failure handling; won 1st place among 100+ teams and validated the platform with 200+ founders.

AY Logistics — Autonomous Vehicle Inspection Platform | Python, FastAPI, PostgreSQL, Computer Vision Mar 2026 – Jul 2026

- Built a computer-vision inspection pipeline using RGB, thermal, and telemetry data to detect vehicle defects and thermal anomalies.
- Evaluated the system across 12,000+ labeled inspection images, achieving 98.4% recall on the project evaluation dataset.
- Integrated inspection workflows with FastAPI and PostgreSQL backend services, reducing inspection time from 45 to 12 minutes and surfacing defects within 3 seconds.

Leadership & Professional Development

Management Leadership for Tomorrow (MLT)

Nov 2025 – Present

Career Prep Fellow

- Selected for an 18-month professional development program focused on technical development, leadership, mentorship, and career preparation.

Additional Involvement: ColorStack (Member); UNL LeetCode Club (President); National Society of Black Engineers